MIL-S-14022A
6 May 1983
SUPERSEDING
MIL-S-14022(Cml)C)
30 March 1955

MILITARY SPECIFICATION

SODIUM PERBORATE, TETRAHYDRATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two grades of sodium perborate tetrahydrate.

1.2 Classification. Sodium perborate shall be of the following grades as specified (see 6.1):

Grade A - Analyzed reagent
Grade B - Technical

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:46Z
Check the source to verify that this is the current version before use.

PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-186 - Containers, Packaging and Packing For Drugs, Chemicals, and Pharmaceuticals
PPP-C-2020 - Chemicals, Liquid, Dry, and Paste: Packaging Of
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193 - Reagent Water
D1921 - Particle Size (Sieve Analysis) of Plastic Materials
E11 - Wire-Cloth Sieves For Testing Purposes

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:46Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Physical.

3.1.1 Form and color. Sodium perborate shall be a white crystalline or granular powder when tested as specified in 4.2.4.1.

3.1.2 Particle size. The particle size of the sodium perborate shall be such that not more than 5.0 percent of the material shall be retained on a 600-micrometer sieve when tested as specified in 4.2.4.2.

3.1.3 Shelf life. Sodium perborate, tetrahydrate has a type I 12-month shelf life and shall be no more than 90 days old upon delivery.

3.2 Chemical.

3.2.1 Grade A. Grade A sodium perborate shall conform to the chemical characteristics of table I when tested as specified therein.

TABLE I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Available oxygen (O)	9.5		4.2.4.3
Chloride (Cl)		0.003	4.2.4.5
Phosphate (PO_4)		0.003	4.2.4.6
Sulfate (SO_4)		0.005	4.2.4.7
Heavy metals (as Pb)		0.002	4.2.4.8
Iron (Fe)		0.001	4.2.4.9

3.2.2 Grade B.

3.2.2.1 Available oxygen. Grade B sodium perborate shall contain not less than 9.5 percent available oxygen when tested as specified in 4.2.4.3.

3.2.2.2 pH of 3-percent solution. The pH of a 3-percent solution of grade B sodium perborate shall be 9.9 to 10.3 when tested as specified in 4.2.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:46Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the sodium perborate of one grade, produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.2).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For sodium perborate test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 110 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for sodium perborate test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:46Z
Check the source to verify that this is the current version before use.

(a) Contents per containers not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Fiberboard pads missing (when required)
(f) Unitization not as specified
(g) Marking incorrect, missing, or illegible

4.2.3.2 For sodium perborate test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

4.2.4 Sodium perborate tests. Water in accordance with ASTM D1193, type as applicable, and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Form and color. Visually examine the specimen for form and color.

4.2.4.2 Particle size. Determine the particle size of the specimen in accordance with ASTM D1921, method B, using a 600-micrometer sieve conforming to ASTM E11, except that a 90 $\pm$ 0.1-g sample shall be used instead of a 100-g sample.

4.2.4.3 Available oxygen. Weigh to the nearest 0.1 milligram (mg), approximately 0.3 g of the specimen and transfer to a 300-milliliter (mL) Erlenmeyer flask. Add about 80 mL of cold water, acidify immediately with 20 mL of 0.1N sulfuric acid and titrate at once with 0.1N potassium permanganate. Calculate the percent by weight available oxygen as follows:

$$\text{Percent available oxygen} = \frac{0.8\ AB}{W}$$

where: A = Volume of potassium permanganate used, in milliliters,
B = Normality of potassium permanganate, and
W = Weight of the specimen, in grams.

4.2.4.4 pH of 3-percent solution. Prepare a 3-percent solution of the specimen by dissolving 3.0 g of the material in 100 mL of water. Determine the pH of the solution with a Beckman or equivalent pH meter.

4.2.4.5 Chloride.

(a) Chloride for control. Dissolve 0.1649 g of sodium chloride, NaCl, in sufficient water to make 1 liter (L). One mL of this solution equals 0.1 mg of chloride, Cl.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:46Z
Check the source to verify that this is the current version before use.

(b) Procedure. Dissolve 5.000 g of the specimen in 50 mL of water with the aid of 5 mL of 70-percent nitric acid. Add 3 mL of a 2-percent aqueous solution of silver nitrate. Compare the turbidity produced with that of a control made with 0.15 mg of chloride [1.5 mL of (a)] in an equal volume of solution and treated with the same quantities of reagents as used in the test. If the turbidity in the specimen test is greater than that in the control, the specimen does not comply with the chloride requirement of table I.

4.2.4.6 Phosphate.

(a) Phosphate for control. Dissolve 0.143 g of monobasic potassium phosphate, KH_2PO_4, in sufficient water to make 1 L of solution. One mL of this solution equals 0.1 mg phosphate, PO_4.

(b) Procedure. Dissolve 1.000 g of the specimen in 10 mL of water. Add 2 mL of hydrochloric acid and evaporate to dryness on a steam bath. Take up in 25 mL of approximately 0.5N sulfuric acid and filter. To the clear filtrate add 1 mL of ammonium molybdate reagent solution (50 g of ammonium molybdate, $(NH_4)_6Mo_7O_{24}\cdot 4H_2O$, in 1 L of 1N sulfuric acid) and 1 mL of p-methylaminophenol sulfate reagent solution (0.2 g of p-methylaminophenol sulfate, $(HOC_6H_4NHCH_3)_2H_2SO_4$, and 20 g sodium bisulfate, $NaHSO_3$, in 100 mL of water) and allow to stand for 2 hours at room temperature. In order to comply with the phosphate requirement of table I, any blue color produced shall not exceed that produced by 0.03 mg of phosphate, PO_4, [0.3 mL of control solution (a)] in an equal volume of solution containing the quantities of reagents used in the test, including the residue from evaporation of 2 mL of hydrochloric acid.

4.2.4.7 Sulfate.

(a) Sulfate for control. Dissolve 0.1814 g of potassium sulfate, K_2SO_4, in sufficient water to make 1 L. One mL of this solution equals 0.1 mg of sulfate, SO_4.

(b) Procedure. Dissolve 2.000 g of the specimen in 50 mL of water. Add 2 mL of 37-percent hydrochloric acid and 2 mL of 12-percent barium chloride solution. Compare the turbidity produced with that in a control made with 0.1 mg of sulfate [1.0 mL of (a)] in an equal volume of solution and treated with the same quantities of reagents as used in the test. If the turbidity in the specimen test is greater than that in the control, the specimen does not comply with sulfate requirement of table I.

4.2.4.8 Heavy metals (as Pb).

(a) Lead for control. Dissolve 0.0160 g of lead nitrate in sufficient water to make 1 L. One mL of this solution equals 0.1 mg of lead, Pb.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:46Z
Check the source to verify that this is the current version before use.

(b) Procedure. Dissolve 2.000 g of the specimen in 25 mL of water. Add 1 mL of 37-percent hydrochloric acid and filter any precipitate that may form. Neutralize the filtrate with 10-percent ammonium hydroxide and dilute to 40 mL. To 10 mL add 0.02 mg of lead [2.0 mL of (a)], dilute to 30 mL and add 1 mL of 1N acetic acid (A). To the remaining 30 mL add 1 mL of 1N acetic acid (B). Then to each add 10 mL of a freshly-prepared, saturated aqueous solution of hydrogen sulfide. If (B) is darker than (A), the specimen does not comply with the heavy metals requirement of table I.

4.2.4.9 Iron.

(a) Iron for control. Dissolve 0.702 g of ferrous ammonium sulfate, $Fe(NH_4)_2(SO_4).6H_2O$ in 10 mL of 10-percent sulfuric acid and dilute to exactly 100 mL. To a 10-mL aliquot of the solution add 10 mL of 10-percent sulfuric acid and dilute to 1 L. One mL of this solution equals 0.01 mg of iron, Fe.

(b) Procedure. Dissolve 1 g of the specimen in 40 mL of water and 3 mL of hydrochloric acid and dilute to 50 mL. Add 30 to 50 mg of ammonium persulfate crystals and 3 mL of ammonium thiocyanate reagent solution (300 g of ammonium thiocyanate, NH_4CNS, in water to make 1 L of solution). In order to comply with the iron requirement of table I any red color produced shall not exceed that produced by 0.01 mg of iron [1 mL of control solution (a)] in an equal volume of solution containing the quantities of reagents used in the test.

4.2.5 Container leakage test. Place the container in each of the following positions, and leave it in each for a period of 15 minutes:

(a) Upright
(b) Upside down
(c) On one side (or one quadrant)
(d) On one end (or second quadrant)
(e) On other side (or fourth quadrant)

Examine the container after each period for any evidence of leakage.

5. PACKAGING

5.1 Preservation. Sodium perborate shall be preserved level A or industrial as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit packing. A 1-pound (lb) quantity (+0.2 or -0 ounce) of sodium perborate shall be unit packed in accordance with the applicable requirements of PPP-C-2020 in an amber colored glass, wide-mouth, screw cap bottle. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:46Z
Check the source to verify that this is the current version before use.

bottle shall conform to PPP-C-186, group A, class 1, type of glass optional, style 2, light penetration grade 1, with screw-cap closure A, B or R and outer seal A. The screw-cap shall be tightened to a torque within a range as specified by the bottle manufacturer. There shall be no evidence of leakage of contents when tested as specified in 4.2.5.

5.1.1.2 Intermediate packing. Twelve 1-lb quantity bottles of sodium perborate shall be intermediately packed upright in a close-fitting fiberboard box. The box shall conform to minimum grade W6c of PPP-B-636, and shall be furnished with full bottle height fiberboard, half-slotted partitions, sidewall liner, and top and bottom pads. The partitions shall form a close-fitting cell for each bottle. Motion of contents shall be prevented by inserting fiberboard pads wherever needed. Partitions, liner, and pads shall be formed of material conforming to minimum grade W6c of PPP-F-320. The box shall be closed as specified in the appendix to PPP-B-636.

5.1.2 Industrial. Sodium perborate shall be unit packed industrially in a 1-lb quantity, in accordance with MIL-STD-1188.

5.2 Packing. Sodium perborate shall be packed level A, B or industrial as specified (see 6.2).

5.2.1 Level A. Two intermediate packs of 1-lb bottles of sodium perborate shall be packed upright in a nailed wood, cleated plywood, or wirebound wood box at the option of the contractor. The intermediate packs shall be arranged in two layers, each consisting of two rows of two boxes in the most compact configuration. The nailed wood box shall conform to class 2, style 4, grade A of PPP-B-621, the cleated plywood box shall conform to overseas type style A or B, grade A of PPP-B-601, and the wirebound wood box shall conform to class 3, style optional of PPP-B-585. All boxes shall be for a type 2 average load. The box of PPP-B-585 shall be preserved in the manner as specified for the grade A box of PPP-B-621. Motion of contents in the box shall be prevented by inserting fiberboard pads where needed. Fiberboard shall conform to minimum grade W6c of PPP-F-320. The box shall be closed and reinforced as specified in the applicable box specification.

5.2.2 Level B. The intermediate packs of 1-lb quantity bottles shall be packed level B in the same manner as specified for those packed level A, except that the box shall conform to grade V3c of PPP-B-636.

5.2.3 Industrial. The 1-lb quantity bottles of sodium perborate shall be packed in accordance with MIL-STD-1188.

5.3 Unitization. Level A and B packs of sodium perborate shall be unitized in accordance with load type I of MIL-STD-147. Industrial packs shall be unitized to assure safe delivery to destination and safe stacking in storage. Loads shall be acceptable to common carriers and shall be in compliance with the Uniform Freight Classification Rules, National Motor Freight Classification Rules and the rules and regulations applicable to any other selected mode of transportation.

Source: https://assist.dla.mil -- Downloaded: 202640-18T02:46Z
Check the source to verify that this is the currentversion before use.

5.4 Marking. Level A and B unit packs, intermediate packs, packs and unitized loads shall be marked in accordance with MIL-STD-129. Industrial unit packs and packs shall be marked in accordance with MIL-STD-1188. Industrial unitized loads shall be marked to assure prompt and safe delivery to destination. Each container and unitized load shall be marked to show that the sodium perborate, tetrahydrate has a type I, 12-month shelf life and to show the date of manufacture and lot or batch number of the sodium perborate. In addition, each container shall be marked as follows:

"Sodium perborate should not be stored at temperatures above 32°C (89.6°F) to reduce loss of active oxygen."

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Grade of material required (see 1.2), and
(c) Level of preservation and packing required (see 5.1 and 5.2).

6.2 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.3 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

Custodian:

Army - EA

Review activities:

Army - MD
DLA - GS

Preparing activity:

Army - EA

Project No. 6810-B383

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:46Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for submitting comments and suggestions for improvements. All users of military standardization documents are invited to p suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)* mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorizati to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO P
NEC
IF M
IN
UNITE

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON, D.C.

POSTAGE WILL BE PAID BY DEPARTMENT OF THE ARMY

Commander
US Army Armament Research
and Development Command
ATTN: DRDAR-TSC-S
Aberdeen Proving Ground, MD 21010

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-14022A	SODIUM PERBORATE, TETRAHYDRATE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

EDITION OF 1 OCT 76 IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:46Z
Check the source to verify that this is the current version before use.